

PRISM (Predictive Risk Insights & Scoring Model)

By

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Introduction to the Project



Basic Profile: Hazel Sequeira, Freia Quadros - B.E. Electronics & Computer Science
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Basic Project Information: PRISM - Predictive Risk Insights & Scoring Model

Our solution is an enhanced credit scoring and risk assessment system that integrates multi-modal data extracted from traditional and non-traditional sources, explainable AI insights, adaptive fraud detection along with document verification and regulatory compliance in one engine.

1. **Multi-modal credit scoring and risk assessment** using traditional credit history along with utility payments and EPFO (Employees Provident Fund Corporation) employment verification for new-to-credit individuals.
2. **Sentiment-augmented portfolio & risk simulation** using NLP techniques such as Topic Modelling to improve decision precision, understand loan amount-based risk stratification, and improve repayment tenure optimization.
3. **Explainable AI** (SHAP + counterfactual what-if analysis) enabling regulator-friendly transparent scoring and delivering better insights for the users to understand and improve their financial well-being.
4. **Adversarial fraud learning & anomaly detection** for loan stacking, income forgeries, tampered documents and synthetic identities.
5. **AI-based document authenticity verification** (metadata, watermark, font, pixel analysis).
6. **Regulatory compliance engine** automating KYC/AML & DPDPA-aligned consent flow.

Borrower Dashboard

Lender Dashboard

Risk Dashboard

Consolidated borrower risk overview

Composite Credit Score

300 Neutral

Score generated using employment consistency, utility repayment behavior, and sentiment signals.

Risk Signals

Low employment history impact ✓

View Score Explanation

Simulate Risk Scenarios

Share with Lender

Risk Dashboard

Consolidated borrower risk overview

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Risk Signals

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View Fraud Analysis

View Compliance Status

Borrower receives an explainable credit score with clear reasons

Lender reviews fraud and risk signals for informed decisions

Score Explanation

A transparent breakdown of the key factors that influenced the generated credit score.

MODEL REASON CODES

Income is below the preferred threshold ✓

Employment tenure is less than one year ✓

Limited credit history available ✓

SYSTEM & COMPLIANCE CHECKS

Fraud Indicators	None Detected
Identity Verification	Verified
Consent Status	Logged & Active

BEHAVIORAL SIGNAL

Applicant intent analysis indicates:

Neutral

This explanation is generated using rule-based explainability aligned with regulatory expectations. Each reason directly maps to a deterministic condition in the scoring engine, ensuring auditability and customer fairness.

Back to Dashboard

Fraud & Anomaly Detection

Automated detection of suspicious patterns, inconsistencies, and adversarial fraud signals during loan assessment.

IDENTITY & DOCUMENT CHECKS

Document Authenticity **Verified**

Identity Consistency **Verified**

BEHAVIOURAL & FINANCIAL ANOMALIES

Application Behaviour **Unusual Frequency**

Sentiment Risk **Neutral**

NETWORK & LOAN STACKING

Loan Stacking Status **FLAGGED**

Detection Reason: Multiple loan applications detected within 24 hours

Fraud Risk Assessment Outcome

Multiple loan applications detected in a short time window. Manual review recommended.

Risk Level: High

Recommendation: Manual Review

Fraud detection is implemented using a frequency-based heuristic that identifies abnormal loan application behaviour. This approach is effective for early-stage risk filtering and can be extended with ML-based anomaly detection in production.

Technical Approach & Feasibility

1. System Architecture - N-tier microservices with decoupled ingestion, analytics & storage

- FastAPI ingestion with PostgreSQL + Redis for low-latency processing
- AES-256 encryption at rest and TLS 1.3 in transit
- DPDPA 2023-compliant data governance

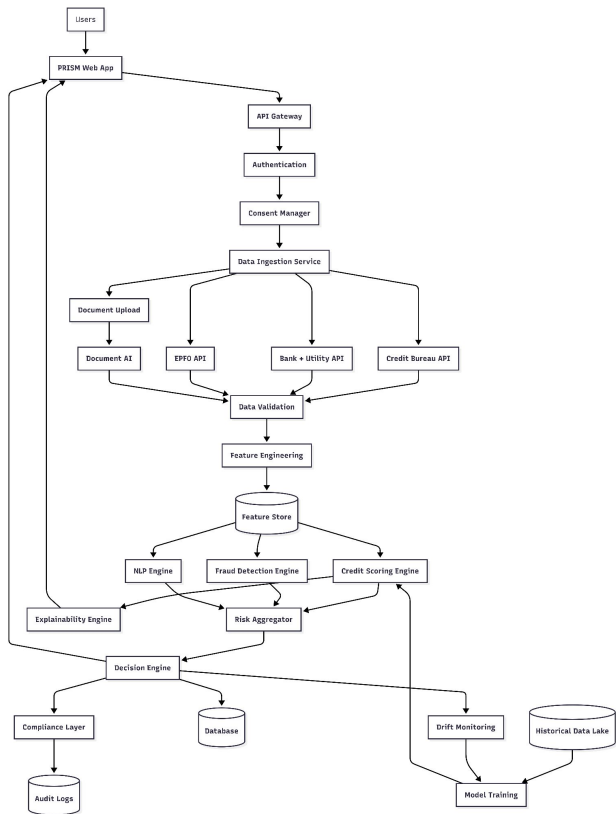
2. Intelligent Risk Pipeline - XGBoost-based credit scoring for complex, non-linear patterns

- WoE/IV-driven feature selection using alternative data (EPFO, utilities, banking)
- NLP-based behavioral signals fused with structured financial features

3. Trust & Fraud Layer - Detection of synthetic identities and fraud networks using Isolation Forests

- Collision fraud and anomaly detection at network level
- SHAP-based explainability with audit-grade regulatory compliance

Area	Feasibility Approach
Technology	Open-source ML stack + microservices
Compliance	Built-in consent, encryption, audit logs
Deployment	Cloud-native containerized architecture



Architecture Diagram

Project Execution Plan and Demo Milestones

Phases	Duration	Focus Area	Monthly Demo Outcomes
1	Mar – Apr 2026	System Foundation	Core system architecture, borrower–lender flow design, and compliance mapping
2	Apr – Mid Jun 2026	Data Preparation	Monthly demos of borrower data input, validation, and structured data handling
3	Mid Jun – Aug 2026	Credit Scoring & Sentiment	Monthly demos showing rule-based credit score generation and sentiment-based risk indicators
4	Aug – Sept 2026	Explainability Module	Clear explanation of score drivers and influencing factors
5	Sept – Oct 2026	Fraud & Document Checks	Monthly demos of loan stacking detection and document authenticity verification
6	Oct – Nov 2026	System Integration	End-to-end borrower-to-lender demo with all modules connected
7	Nov – Dec 2026	Final Testing & Documentation	System refinement, testing, documentation, and final project demo